FUNCTIONAL SPECIFICATION DOCUMENT

Moji Desk Buddy

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1. Introduction :

This document presents the functional specification of the Desk Buddy system named MOJI.

The purpose of this document is to define the functional behavior of the system, describe its internal subcomponents, and explain how each functionality operates.

2. System Overview :

MOJI is an intelligent desk assistant designed to interact with users through audio and visual inputs. The system processes user commands, detects presence, analyzes emotions, and manages tasks and reminders.

3. Functional Subcomponents Description :

i. Audio Input Component :

- **Description :**

The Audio Input Component captures user speech and transforms it into structured command data.

- **Functional Behavior :**

  - Detects wake word activation.
  - Converts speech into text.
  - Recognizes user intent.
  - Classifies the command type.

- **Output :**

Structured command data routed to the corresponding functionality:

  - Task Management
  - Reminder Management
  - Decision Support

ii. Audio Output Component :

- **Description :**

The Audio Output Component generates spoken responses based on processed system data.

- **Functional Behavior :**

  - Formats response text.
  - Converts text into speech.
  - Outputs audible feedback to the user.

iii. Human Presence Detection Component :

- **Description :**

This component verifies whether a user is physically present in front of MOJI.

- **Functional Behavior :**
  - Captures visual frames.
  - Performs face detection.
  - Confirms human presence.
  - Updates system interaction state.
- **Output :**

Presence status:

  - User Detected
  - No User Detected

Additional Output (Visual Feedback via OLED Display):

When a user is detected, the system may activate visual expressions on the OLED display to indicate system awareness (e.g., eye movement or active state animation).

When no user is detected, the OLED display may switch to idle mode or display neutral static eyes.

This status enables or disables interaction functionalities.

iv. Emotion Detection Component:

- **Description :**

The Emotion detection Component classifies the emotional state of the detected user.

- **Functional Behavior :**

  - Extracts facial features.
  - Applies emotion classification logic.
  - Generates emotion label output.

- **Output :**

Emotion state (e.g., Happy, Neutral, Stressed).

This information is used to adapt:

  - Reminder tone
  - Recommendation style
  - Response behaviour

Additional Output (OLED Emotional Expression):

The detected emotional state may also be represented visually through the OLED display using animated eye expressions or facial patterns corresponding to the classified emotion.

For example:

  - Happy → smiling or curved-eye animation
  - Sad → drooping-eye animation

This visual representation enhances human-robot interaction by providing non-verbal feedback.

v. Task Management Component:

- **Description :**

The Task Management Component allows the creation and storage of user tasks.

- **Functional Behavior :**

  - Extracts task details from structured command.
  - Validates task data.
  - Stores task in local storage.
  - Updates internal task list.
  - Generates confirmation message.

vi. Reminder Management Component :

- **Description :**

This component is responsible for monitoring scheduled tasks and triggering reminders.

- **Functional Behavior :**
  - Continuously monitors system time.
  - Matches stored tasks with current time.
  - Generates reminder notification.
  - Allows enabling/disabling reminder feature.

Reminders may adapt according to detected emotional state.

vii. Decision Support Component :

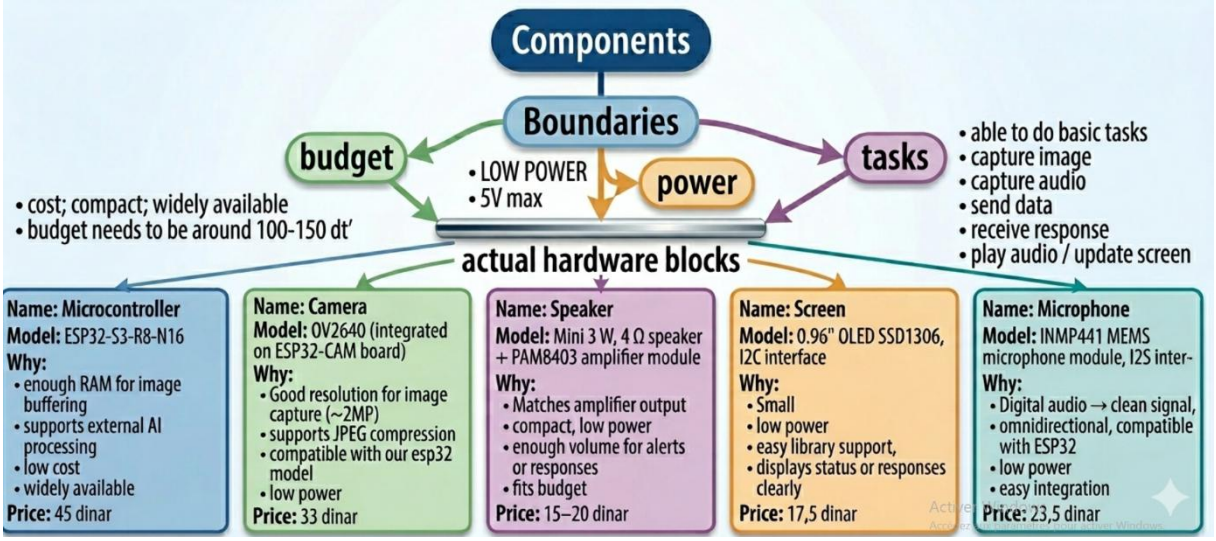
- **Description :**

This component provides intelligent task recommendations.

- **Functional Behavior :**
  - Retrieves pending tasks.
  - Analyzes deadlines and priority levels.
  - Incorporates emotional context.
  - Generates recommendation output.

The final output is delivered through audio response..

4. Budget Estimation :

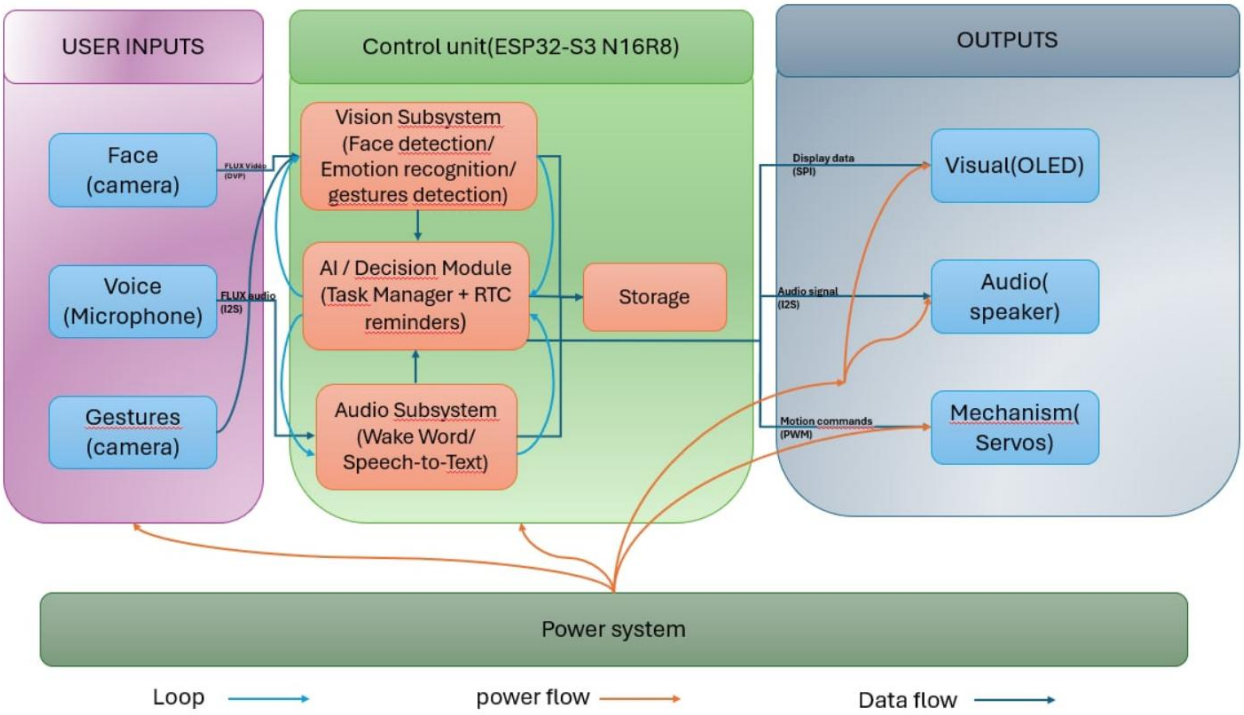


Component (model name)	Reason	Link
esp32S3R8N16	enough RAM for image buffering supports external AI processing low cost	Carte de developpment ESP32-S3 double USB-C WROOM-1 N16R8
OV2640 (integrated on ESP32-CAM board)	resolution for image capture (~2MP) supports JPEG com compatible with our esp32 model	ESP32-CAM Micro USB vers Port série avec Module de caméra OV2640 2M
Model: 0.96" OLED SSD1306, I2C interface	Small Low power easy library support	INMP441 Module microphone omnidirectionnel MEMS Interface I2S Supporte ESP32
Model: Mini 3 W, 4 Ω speaker + PAM8403 amplifier module	Matches amplifier output compact Lowpower enough volume for alerts or responses	Module PAM8403 2x3W Amplificateur Audio 5V
INMP441 MEMS microphone module	Omnidirectional compatible with ESP32 lowpower easy integration	Haut-parleur 3070 4 Ohms 3W – Son puissant et compact pour projets électroniques | ELECTRONIC SHOP

5. Diagram References:

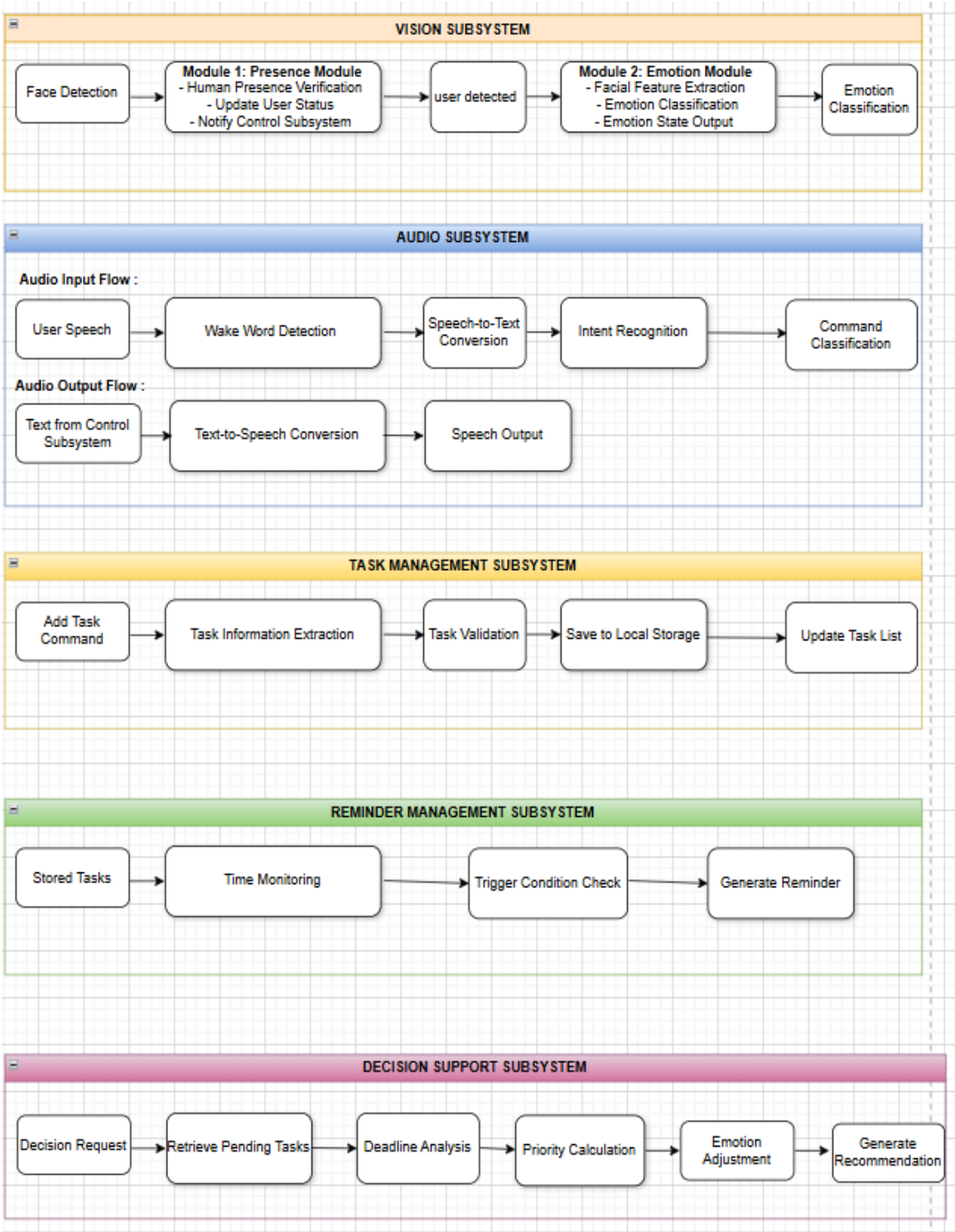
The system architecture is supported by the following diagrams:

i. Global Block Diagram :



ii. Mind Map :

iii. Subsystems Architecture Diagram :



6. Conclusion :

The functional specification of MOJI defines the internal organization of the system through clearly separated subcomponents, each performing a precise and well-defined role.

The system combines audio processing, visual perception, task management, reminder monitoring, decision support, and expressive feedback mechanisms to ensure coherent and intelligent interaction.

Each functional module operates independently while contributing to the overall system behavior, ensuring modularity, scalability, and structured development.This specification serves as a reference framework for implementation, testing, and system validation.